

A Comparative Machine Learning Approach for Predicting Cricket Bowling Performance Using Statistical Feature Analysis

Abstract

This study presents a machine learning-based framework to predict cricket bowling performance using statistical data. Multiple algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbors were implemented. The models were evaluated using accuracy, precision, recall, and F1-score. Results indicate that Random Forest achieved the highest accuracy. Techniques such as cross-validation and hyperparameter tuning improved model performance, demonstrating the effectiveness of machine learning in sports analytics.

1. Introduction

The use of machine learning in sports analytics has grown significantly. Cricket generates large datasets that can be used to analyze player performance. Traditional statistical methods are limited, while machine learning models can capture complex relationships. This study aims to predict bowling performance using ML techniques.

2. Dataset Description

The dataset contains cricket bowling statistics such as overs bowled, economy rate, and match details. A target variable was created by classifying performance into Good, Average, and Poor categories based on economy rate.

3. Methodology

Data preprocessing included handling missing values and encoding categorical variables. The dataset was split into training (75%) and testing (25%). Multiple machine learning models were applied and evaluated using standard metrics.

4. Results and Analysis

Random Forest achieved the highest accuracy compared to other models due to its ensemble learning capability. SVM also performed well, while Logistic Regression showed lower accuracy.

5. Accuracy Improvement Techniques

Cross-validation, feature selection, and hyperparameter tuning were applied to improve model performance and reduce overfitting.

6. Conclusion

Machine learning models effectively predict cricket bowling performance. Random Forest was the best-performing model. Future work includes deep learning and real-time analytics.